

From: "Policy & Information Team, Newcastle University"

Subject: Ethics Form Completed for Project: Comprehensive Evaluation and Benchmarking of Small Language Models for Edge Computing Tomasz Szydło

Date: Thu, 28 May 2026 21:32:44 +0200

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Ref: 77803/2026

Thank you for submitting the ethical approval form for the project 'Comprehensive Evaluation and Benchmarking of Small Language Models for Edge Computing ' (Lead Investigator: Tomasz Szydło). Expected to run from 01/06/2026 to 13/08/2026.

Based on your answers, the University Ethics Committee grants its approval for you to start working on your project. Please be aware that if you make any significant changes to your proposal then you should complete this form again, as further review may be required. This confirmation may be used within a research portfolio as evidence of ethical approval. Please note: this confirmation will be the only correspondence you should expect to receive as evidence of ethical approval. There will be no other confirmation provided. You may now proceed with research. If you have any queries, please review the internal and external ethics FAQ pages before contacting res.policy@ncl.ac.uk.

Best wishes

Research Policy Intelligence and Ethics Team,

Research Strategy & Development
res.policy@ncl.ac.uk

The University must ensure that all its projects undergo appropriate ethical review before commencement. The University has a two-stage approach, requiring ALL projects to undergo a preliminary ethical assessment process. Further guidance can be found at:

<https://www.ncl.ac.uk/research/research-governance/ethics/>

This form is a reduced version of the University's Preliminary Ethical Assessment Form intended for students taking the module CSC8099. It is assumed that students on these modules will not be operating in a clinical setting; **if you are conducting your study in a clinical setting then you will need to complete the full version on this form** (please contact the module leader).

Please complete the following (reduced) Preliminary Ethical Assessment Form.

SECTION 1: Applicant Details

Name of Researcher (Applicant):	Runnan Feng
Email Address:	c4002159@newcastle.ac.uk

SECTION 2: Project Details

Project Title:	Comprehensive Evaluation and Benchmarking of Small Language Models for Edge Computing
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SECTION 5: Human Participants in a Non-Clinical Setting

Does the research involve human participants (e.g. use of questionnaires, focus groups or observation)?	YES ☐	NO R
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*If you answered **NO** to Section 5, please go directly to the Declaration in Section 7.*

*If you answered **YES** to Section 5, please complete Section 6.*

SECTION 6: Human Participants in a Non-Clinical Setting – Further Information

	YES	NO
1. Does the study involve other vulnerable groups (e.g. children, those with cognitive impairment, or those in unequal relationships (e.g. your own students))?	☐	☐
2. Will the study require the co-operation of a gatekeeper for initial access to the groups or individuals to be recruited (e.g. students at school, members of a self-help group, and residents of a nursing home)?	☐	☐
3. Will it be necessary for participants to take part in the study without their knowledge and consent at times (e.g. covert observation of people in non-public places)?	☐	☐
4. Will this programme/project involve deliberately misleading participants in any way?	☐	☐
5. Will the study involve discussion of sensitive topics (e.g. sexual activity, drug use)?	☐	☐

6. Are any drugs, placebos or other substances (e.g. food substances, vitamins) to be administered to the study participants or will the study involve invasive, intrusive or potentially harmful procedures of any kind?*	☐	☐
7. Will blood or tissue samples be obtained from subjects?	☐	☐
8. Is pain or more than mild discomfort likely to result from the study?	☐	☐
9. Could the study induce psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life?	☐	☐
10. Will the study involve prolonged or repetitive testing?	☐	☐
11. Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?	☐	☐

* **Please Note:** Depending on the details of this project, this may require NHS approval.

*If you answered **NO** to all of the questions in Section 6: Your project does not require Full Ethical Approval. Please go to the Declaration in Section 7.*

*If you have answered **YES** to any of questions in Section 6: You will need to describe more fully how you plan to deal with the ethical issues raised by your research by completing the Full Ethical Approval application form. **This does not mean that you cannot do the research - only that you may need to seek and satisfy an ethical opinion from a Faculty Research Ethics Committee.***

SECTION 7: Declaration

I certify that the information contained in this application is accurate.	
Name of student:	Runnan Feng
Date:	29/05/2026
Name of supervisor:	Tomasz Szydlo

University Ethics Form Version 3

Applicant Details (922)

Is this approval for a: (11240)

Type: (!/list-dropdown)

A2 - Student Project

What type of degree programme is being studied? (11319)

Type: (!/list-dropdown)

A2 - Postgraduate Taught (e.g. MSc / MA)

Name of Principal Researcher (11241)

Type: (S/text-short)

Tomasz Szydło

Please enter your email address (11258)

Type: (S/text-short)

c4002159@newcastle.ac.uk

Please select your school/academic unit (11242)

Type: (!/list-dropdown)

A1 - School of Computing

Please enter the module code (11243)

Type: (S/text-short)

CSC8099

Please enter your supervisor's email (11259)

Type: (S/text-short)

tomasz.szydlo@newcastle.ac.uk

Please select your supervisor's school/unit: (11244)

Type: (!/list-dropdown)

A1 - School of Computing

Project Details (923)

Project Title (11245)

Type: (S/text-short)

Comprehensive Evaluation and Benchmarking of Small Language Models for Edge Computing

Project Synopsis (11257)

Type: (T/text-long)

Small Language Models (SLMs) are compact AI models designed to run locally on resource-constrained hardware without relying on cloud connectivity. They are considered promising candidates for deployment on low-power computing platforms such as embedded systems and single-board computers.

This project conducts a comprehensive evaluation and benchmarking of multiple SLMs across a range of parameter sizes, assessing whether they can deliver acceptable inference quality while satisfying the performance, memory, and energy constraints typical of resource-constrained local deployment.

Models are evaluated using standard benchmark datasets covering diverse cognitive tasks including knowledge reasoning, mathematical reasoning, and commonsense understanding. Experiments are conducted under multiple workload scenarios designed to simulate realistic usage conditions, from normal single-user interaction to sustained and concurrent request patterns.

Key measurement dimensions include output quality, inference speed, memory usage, energy consumption, and performance stability under load. Results are compared against published large language model baselines to quantify the quality trade-off of using smaller, locally-deployed models.

The study aims to identify the optimal balance between model capability and resource efficiency, and to provide evidence-based recommendations for practitioners considering on-device AI solutions as alternatives to cloud-dependent large language models.

Project start date (11260)

Type: (D/date)

01/06/2026

Project end date (11261)

Type: (D/date)

13/08/2026

Is the project externally funded? (11262)

Type: (!/list-dropdown)

A3 - No

Does your project involve collaborators outside of the University? (11265)

Type: (Y/yes-no)

No

Existing Ethics, Sponsorship & Responsibility (930)

Has ethical approval to cover this proposal already been obtained? (11267)

Type: (Y/yes-no)

No

Will anyone be acting as sponsor under the NHS Research Governance

Framework for Health and Social Care? (11270)

Type: (Y/yes-no)

No

Do you have a Newcastle upon Tyne Hospitals (NUTH) reference? (11272)

Type: (Y/yes-no)

No

Will someone other than you (the principal investigator) or your supervisor (for student projects) be responsible for the conduct, management and design of the research? (11274)

Type: (Y/yes-no)

No

Animals (I) (924)

Please note that this question changed on 01/06/25 to capture activity with all animals

Does your proposed research involve the use of non-human animals i.e. the observation, capture or manipulation of animals OR their tissues (i.e. materials from animal such as skin, blood, bone, and other bodily fluids)?

(11246)

Type: (Y/yes-no)

No

NHS, Health & Social Care: Facilities, Staff & Patients (I) (925)

Will the study involve participants recruited by virtue of being NHS patients or service users, their dependents, their carers or human tissues or the use of NHS & Health/Social Care Facilities or otherwise require REC approval? (11247)

Type: (Y/yes-no)

No

Human Participants in a Non-Clinical Setting (I) (926)

Does the research involve human participants e.g. use of questionnaires, focus groups, observation, surveys or lab-based studies involving human participants?

(11249)

Type: (Y/yes-no)

No

Data (I) (927)

Does the research involve the viewing, usage or transfer of sensitive data or personal data as defined by the [General Data Protection Regulation \(GDPR\)](#) or data governed by statute such as the [Official Secrets Act 1989](#) / [Terrorism Act 2006](#) , commercial contract or by convention e.g. client confidentiality? (If you are unsure please tick YES and complete the sub-questions). (11251)

Type: (Y/yes-no)

No

Environment and Responsible Research Innovation (I) (928)

Will the study potentially cause a direct or indirect negative impact on local communities, cultural heritage or the environment? (11253)

Type: (Y/yes-no)

No

International Projects (I) (929)

Will the research be conducted outside of the UK or [European Economic Area \(EEA\)](#) , or will it involve international collaborators outside the EEA? (11255)

Type: (Y/yes-no)

No

Next Steps (931)

Based on your responses your project has been categorised as (ethically) low risk and no further review is required before you start work. You will receive a formal approval email on submission of this form. Should your project change you may need to apply for new ethical approval. (11282)

Type: (X/boilerplate)

Supporting Documentation (940)

**Please upload any documents (not uploaded elsewhere in the application)
which you think are relevant to the consideration of your application. (11308)**

Type: (/upload-files)

Summary and Submission (941)

Thank you for completing the University's Ethical Review Form. Based on your answers the University is satisfied that your project has met its ethical expectations and grants its ethical approval.

Please be aware that if you make any significant changes to your project then you should complete this form again as further review may be required. Confirmation of this decision will be emailed to you.

Please complete the declaration to submit your application.

Declaration

I certify that:

(11314)

Type: (M/multiple-opt)

the information contained within this application is accurate. (11441)

the research will be undertaken in line with all appropriate, University, legal and local standards and regulations. (11442)

I have attempted to identify the risks that may arise in conducting this research and acknowledge my obligation to (and rights of) any participants. (11443)

no work will begin until all appropriate permissions are in place. (11444)

Projects with low risks. All. Confirmation email.

(11338)

Type: (S/text-short)

Based on your answers, the University Ethics Committee grants its approval for you to start working on your project. Please be aware that if you make any significant changes to your proposal then you should complete this form again, as further review may be required. This confirmation may be used within a research portfolio as evidence of ethical approval. Please note: this confirmation will be the only correspondence you should expect to receive as evidence of ethical approval. There will be no other confirmation provided. You may now proceed with research. If you have any queries, please review the internal and external ethics FAQ pages before contacting res.policy@ncl.ac.uk.